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“Analyzing Trust And Ethical Concerns Of AI In
Healthcare Systems”

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1. Executive Summary / Abstract

Artificial Intelligence (AI) is becoming an important part of modern healthcare systems. Hospitals and healthcare organizations are using AI tools for disease diagnosis, patient monitoring, medical imaging, treatment suggestions, and data management. Although AI offers many advantages, there are also serious concerns related to trust, ethics, privacy, fairness, and accountability.

This report analyzes trust and ethical concerns of AI in healthcare systems using survey responses collected from different people including students, patients, healthcare staff, doctors, nurses, and members of the general public. The survey results show that many people believe AI can improve healthcare quality, but they still have concerns about data privacy, incorrect diagnosis, transparency, and over-dependence on technology.

The report also discusses ethical concerns using IEEE, ACM, and BCS codes of ethics. Important issues such as patient privacy, bias in AI systems, lack of transparency, and responsibility for medical mistakes are explained in detail. Legal and regulatory issues including data protection and cyber laws are also discussed.

The report concludes that AI should be used as a support tool for healthcare professionals instead of replacing doctors completely. Strong regulations, regular testing, transparency, and human supervision are necessary to build trust and make AI systems safer in healthcare.

2. Introduction

Artificial Intelligence is changing many industries around the world, and healthcare is one of the most important fields affected by this technology. AI systems are now being used to help doctors diagnose diseases, analyze medical images, recommend treatments, and manage patient records. These systems can work faster and process large amounts of data more efficiently than humans.

The use of AI in healthcare became more common after improvements in machine learning, data science, and cloud computing. During recent years, hospitals and medical

organizations have started using AI-based tools to improve healthcare quality and reduce workload on healthcare staff.

Even though AI provides many benefits, people still have concerns about trusting these systems. Patients may fear that AI can make incorrect decisions or misuse personal medical data. Doctors may worry that AI systems can reduce human judgment or create dependency on machines. These concerns make ethics and trust very important topics in healthcare technology.

This report focuses on analyzing trust and ethical concerns of AI in healthcare systems. The main objectives of this report are:

- To understand public opinion about AI in healthcare.
- To identify major ethical and trust-related concerns.
- To analyze survey results related to AI healthcare systems.
- To apply IEEE, ACM, and BCS ethical principles.
- To suggest recommendations for safer and more trusted AI systems.

3. Problem Statement / Case Background

Artificial Intelligence is becoming increasingly common in healthcare systems around the world. Hospitals and medical organizations now use AI for disease prediction, patient monitoring, medical imaging, virtual assistance, and treatment recommendations. Although these systems improve efficiency, they also create serious ethical and trust-related concerns.

A real-world example that highlights these concerns is the use of AI during the COVID-19 pandemic. Many healthcare organizations introduced AI tools to predict patient conditions and manage hospital resources. However, researchers later discovered that some AI systems produced biased or inaccurate results because the training data was incomplete. In some cases, AI systems failed to perform equally for different patient groups.

Another important case involved healthcare data privacy breaches. Large healthcare databases containing sensitive patient information were targeted by cyberattacks. These

incidents increased public fear regarding AI systems because AI relies heavily on patient data collection and analysis.

These real-world cases show that healthcare AI systems are not only technical tools but also ethical and social responsibilities.

Main Problems Identified

The following major problems were identified during this study:

- Lack of transparency in AI decision-making
- Fear of incorrect diagnosis
- Risk of biased healthcare recommendations
- Privacy and data security concerns
- Confusion regarding responsibility for AI mistakes
- Overdependence on technology in critical medical situations

Stakeholders Involved

Several stakeholders are affected by AI healthcare systems:

- Patients
- Doctors and nurses
- Healthcare staff
- Hospitals and healthcare organizations
- AI developers and software companies
- Government and regulatory authorities
- Society and the general public

Why This Topic Is Important

Healthcare decisions directly affect human life. Even a small mistake can create serious health risks. Therefore, AI systems in healthcare must be accurate, transparent, ethical, and secure. This topic is highly relevant to the computing profession because software developers and computing professionals are responsible for designing systems that protect public safety and human rights.

The survey results collected in this report also show that people support AI in healthcare, but they still want strong human supervision, privacy protection, and ethical regulation.

4. Literature Review / Theoretical Framework

Many researchers believe that Artificial Intelligence can improve healthcare by increasing speed, efficiency, and diagnostic accuracy. AI systems can analyze large amounts of medical data quickly and help doctors identify diseases earlier.

However, researchers also warn that AI systems can create ethical and social problems if they are not properly monitored.

Sara Baase explained in “A Gift of Fire” that computing technologies can create serious ethical risks when public safety and privacy are ignored. Deborah Johnson also discussed that computing professionals must take responsibility for the social impact of technology.

Modern healthcare AI systems depend heavily on machine learning algorithms. These algorithms learn from historical medical data. If the training data contains bias or incomplete information, the AI system may produce unfair results.

Researchers studying AI bias found that some healthcare AI systems performed differently for patients based on gender, ethnicity, or economic background. This creates fairness and discrimination concerns.

Another major issue discussed in previous studies is the “black box” nature of AI systems. Many AI tools provide decisions without clearly explaining how those decisions were reached. This lack of explainability reduces trust among doctors and patients.

Ethical Frameworks

ACM Code of Ethics

The ACM Code states that computing professionals should:

- Avoid harm

- Respect privacy
- Maintain fairness
- Be honest and trustworthy

These principles directly apply to healthcare AI systems because patient safety and privacy are critical.

IEEE Code of Ethics

The IEEE Code emphasizes:

- Public safety
- Transparency
- Responsibility
- Ethical decision-making

Healthcare AI systems must follow these principles because incorrect decisions can harm human life.

BCS Code of Conduct

The BCS Code focuses on:

- Public interest
- Professional competence
- Integrity
- Responsibility toward society

This framework is important because healthcare technology should always prioritize patient welfare over business profit.

Critical Analysis of Existing Research

Most studies focus heavily on the technical benefits of AI but give less attention to long-term ethical consequences.

Although AI can improve efficiency, overdependence on automation may reduce human judgment in healthcare. Some researchers argue that replacing human interaction with AI systems may negatively affect patient trust and emotional support.

Another critical issue is accountability. Current research still does not clearly define who should be responsible when AI systems make harmful medical decisions.

Therefore, healthcare AI systems require strong ethical governance, legal regulations, and continuous monitoring.

5. Methodology / Approach

This report uses both primary and secondary research methods.

Primary Research

A survey was conducted to collect opinions about AI in healthcare systems. Responses were collected from different groups including:

- Students
- Patients
- Doctors and nurses
- Healthcare staff
- General public

More than 100 responses were collected through an online questionnaire.

The survey included questions related to:

- Trust in AI diagnosis
- Transparency of AI systems
- Data privacy concerns
- Acceptance of AI-based treatment
- Responsibility for AI mistakes
- Ethical concerns in healthcare AI

Secondary Research

Books, research articles, ethical codes, and online resources were studied to understand ethical and legal issues related to AI in healthcare.

Tools Used

- Google Forms for survey collection
- Microsoft Excel for charts and data analysis
- Research articles and academic references

The collected data was analyzed using charts and percentage-based comparisons.

6. Analysis and Discussion

The analysis section is based on survey responses collected from participants including students, patients, healthcare staff, doctors, nurses, and members of the general public. The collected data was analyzed using charts and percentage comparisons to understand trust and ethical concerns related to AI in healthcare systems.

The survey responses show that people are interested in the use of AI in healthcare because it can improve efficiency, reduce workload, and support faster diagnosis. However, many participants also expressed concerns regarding privacy, security, transparency, and overdependence on technology.

The following analysis highlights the major findings from the survey.

Survey Findings Overview

Survey Area	Main Finding
Trust in AI	Most participants trust AI only with doctor supervision
Privacy Concerns	Many respondents are worried about misuse of patient data
Transparency	Participants want AI systems to explain decisions clearly
AI-Based Treatment	Most people prefer human confirmation before treatment
Responsibility for Errors	Opinions were divided between doctors, hospitals, and developers

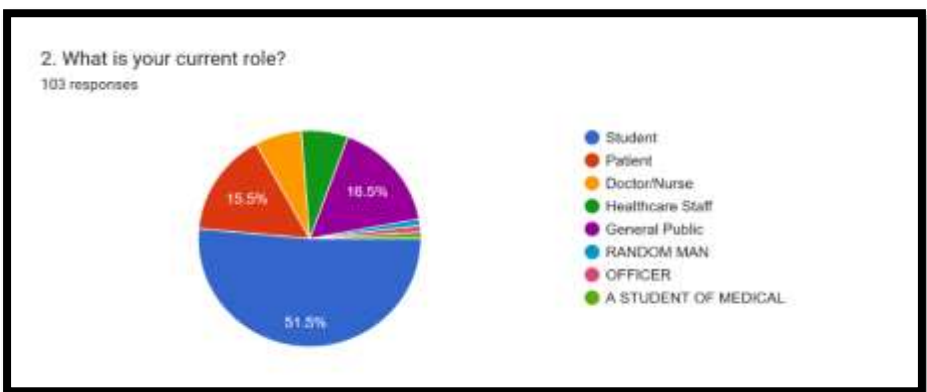
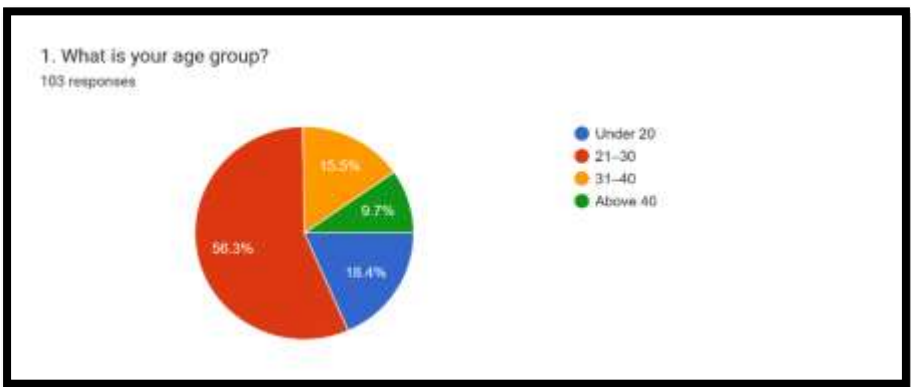
Building Trust	Regular testing and monitoring were considered very important
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The survey responses provided important insights into public trust and ethical concerns related to AI in healthcare.

Demographic Analysis

Most survey participants belonged to the age group of 21–30 years. Students formed the largest group of respondents, followed by members of the general public and patients.

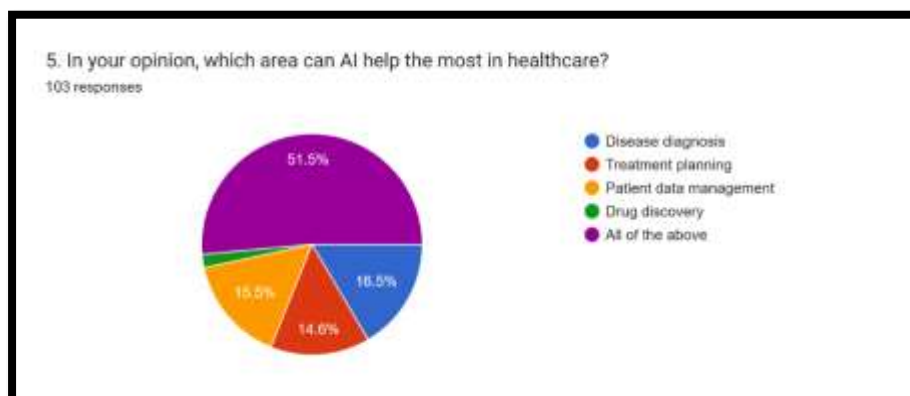
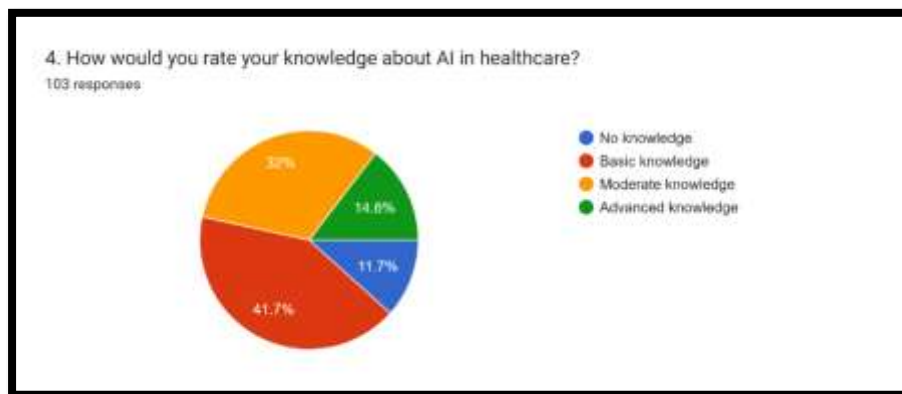
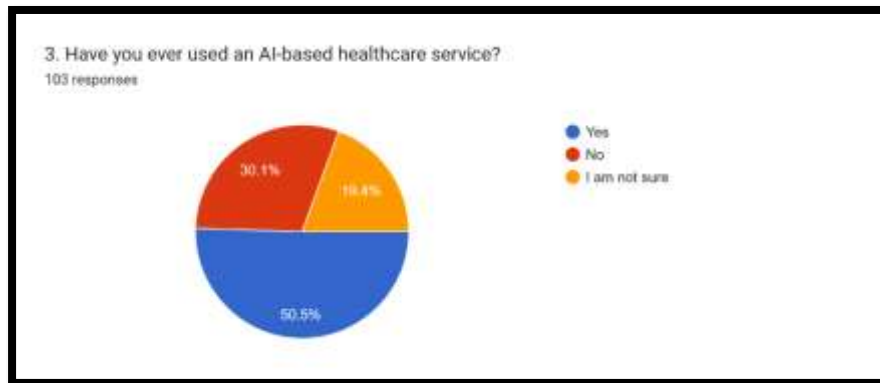
This shows that young people are highly interested in AI technologies and healthcare innovation.

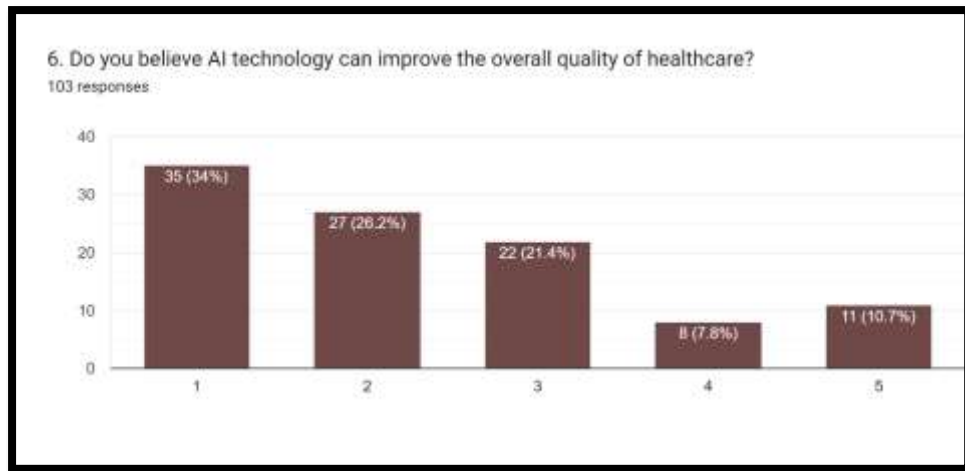


Awareness About AI in Healthcare

Many respondents reported moderate knowledge about AI in healthcare. Some participants had advanced knowledge, while others had only basic understanding.

This result shows that awareness about healthcare AI is increasing, but there is still a need for education and training.

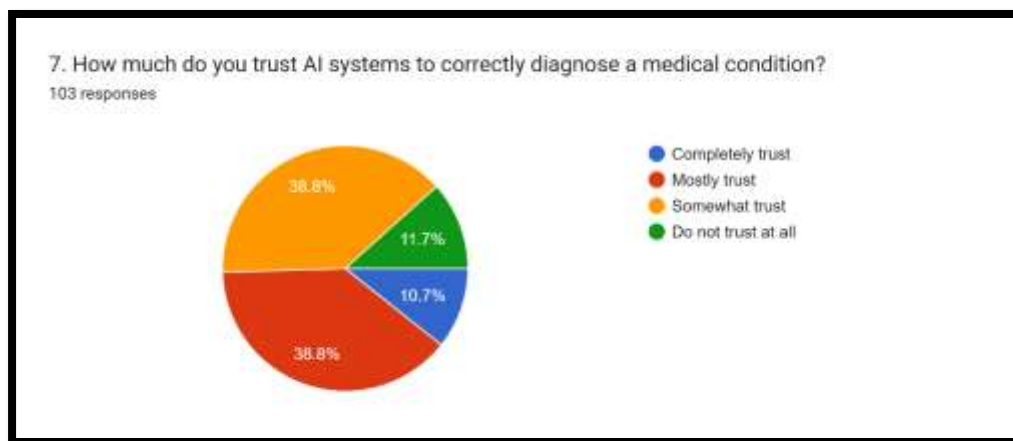




Trust in AI Systems

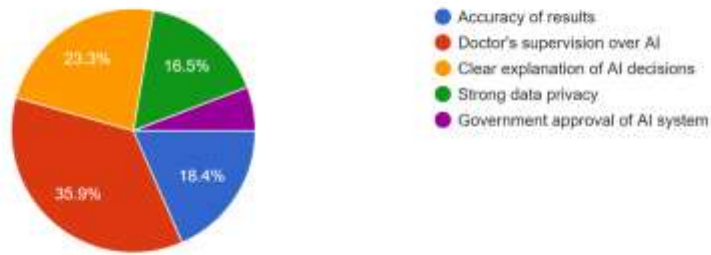
A large number of respondents said they “mostly trust” or “somewhat trust” AI systems for medical diagnosis. However, very few people completely trusted AI systems without human supervision.

This means people are open to AI technology, but they still want doctors to remain involved in decision-making.



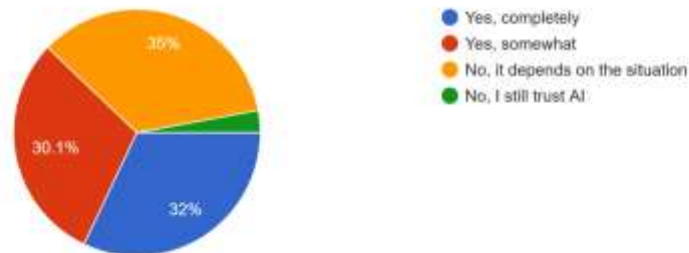
10. Which factor affects your trust in AI healthcare systems the most?

103 responses



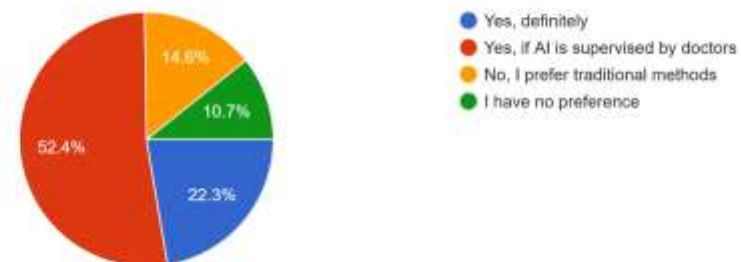
11. If an AI system gave a wrong diagnosis that harmed a patient, would it reduce your trust in all AI healthcare systems?

103 responses



18. Would you choose a hospital that uses AI-assisted diagnosis over one that does not?

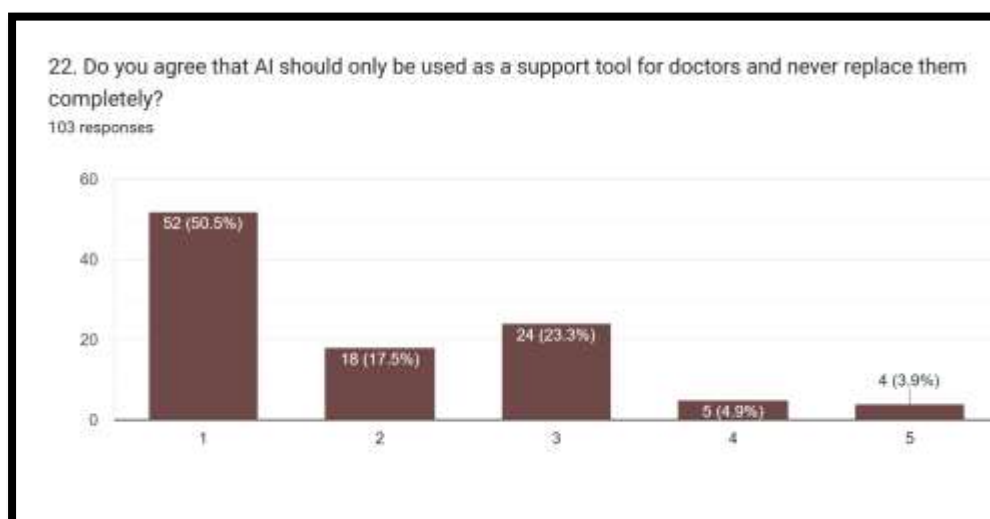
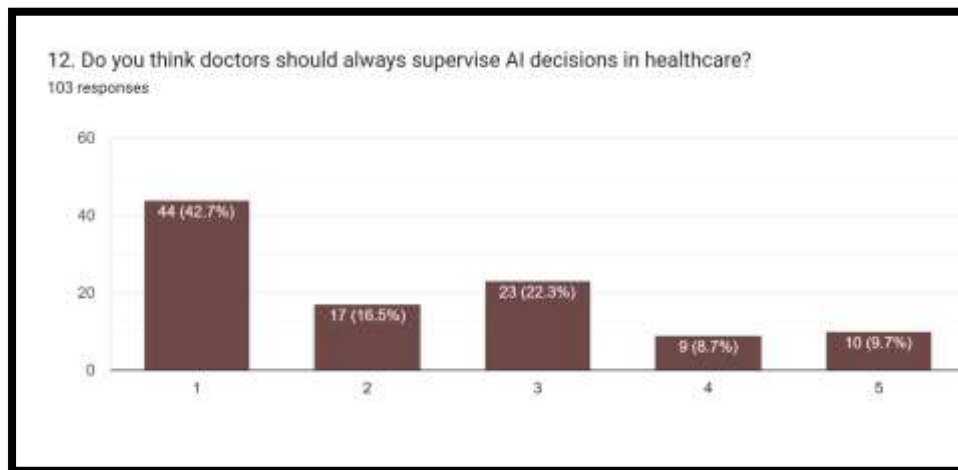
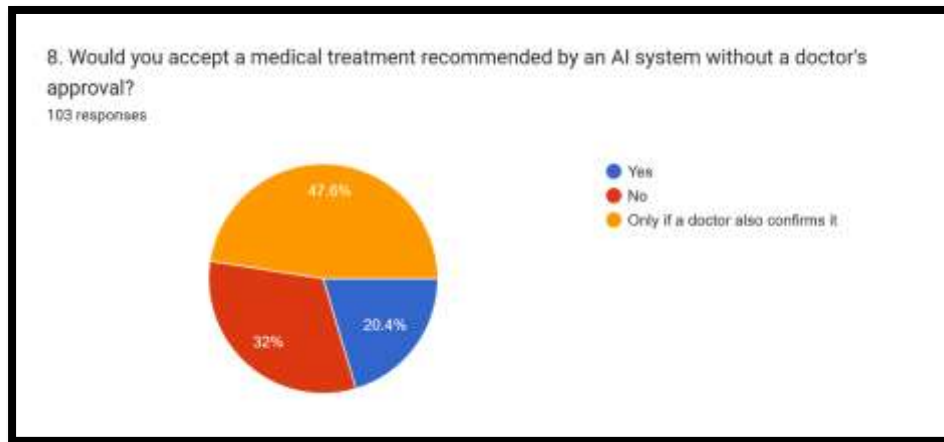
103 responses



Acceptance of AI-Based Treatment

Most participants said they would only accept AI-recommended treatment if a doctor also confirmed it.

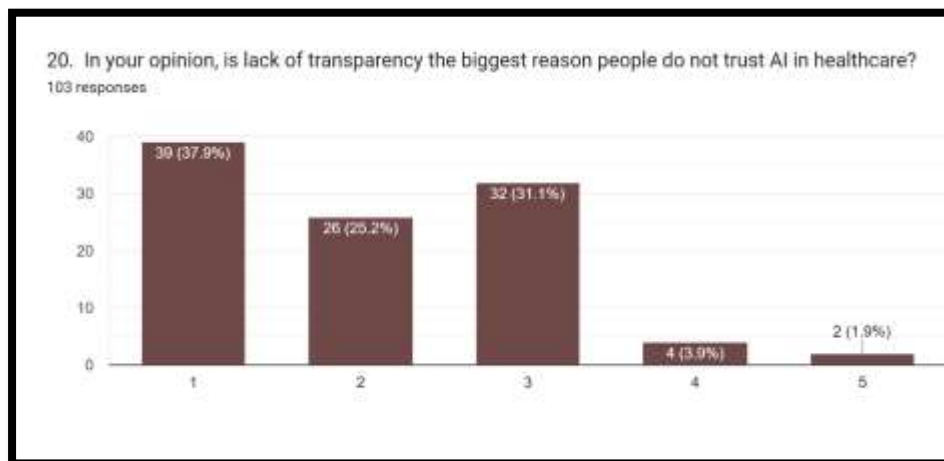
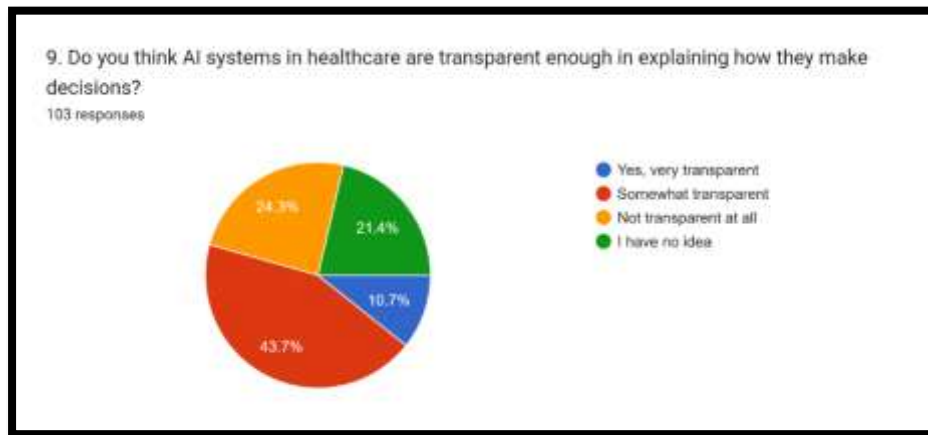
This finding strongly suggests that people prefer AI as a support tool instead of a replacement for healthcare professionals.



Transparency Concerns

Many respondents believed that AI systems are only somewhat transparent. People want clear explanations about how AI systems make decisions.

Lack of transparency reduces trust and creates fear among users.

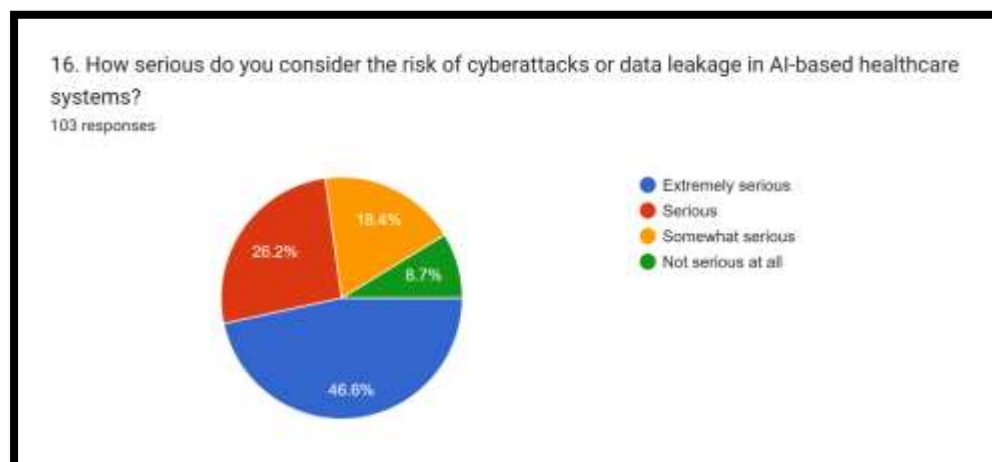
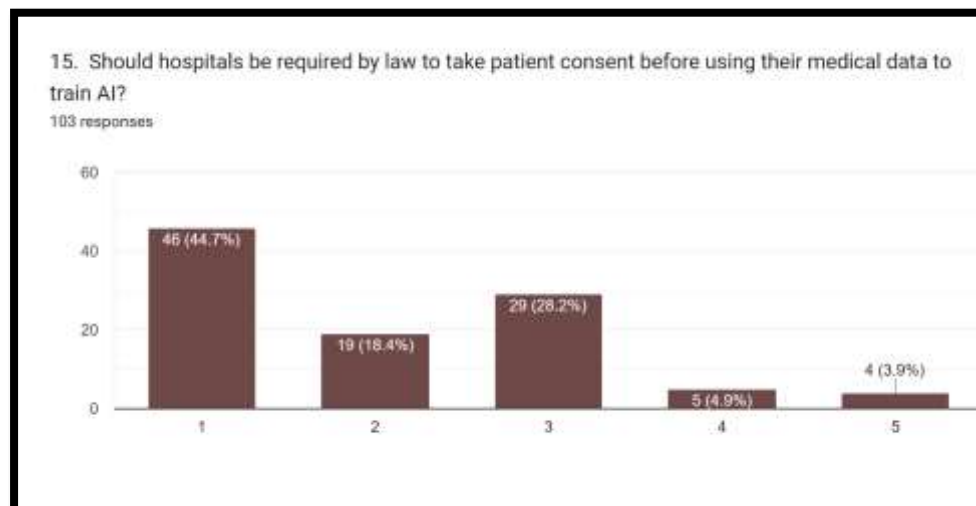
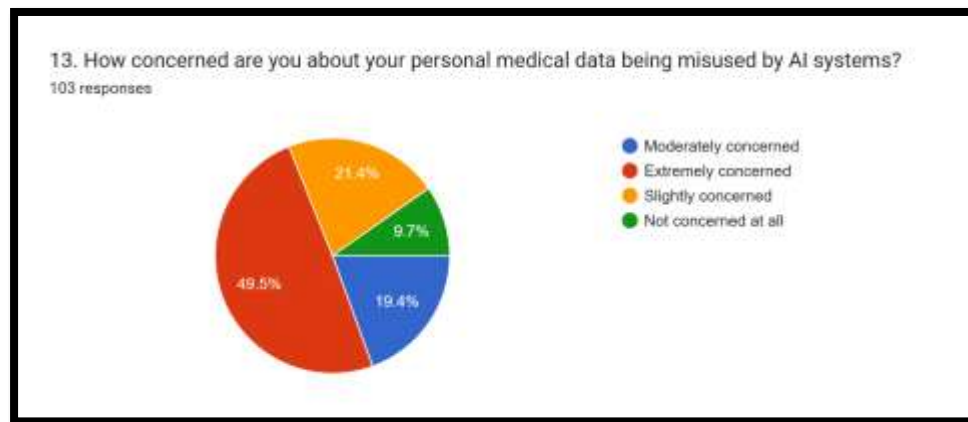


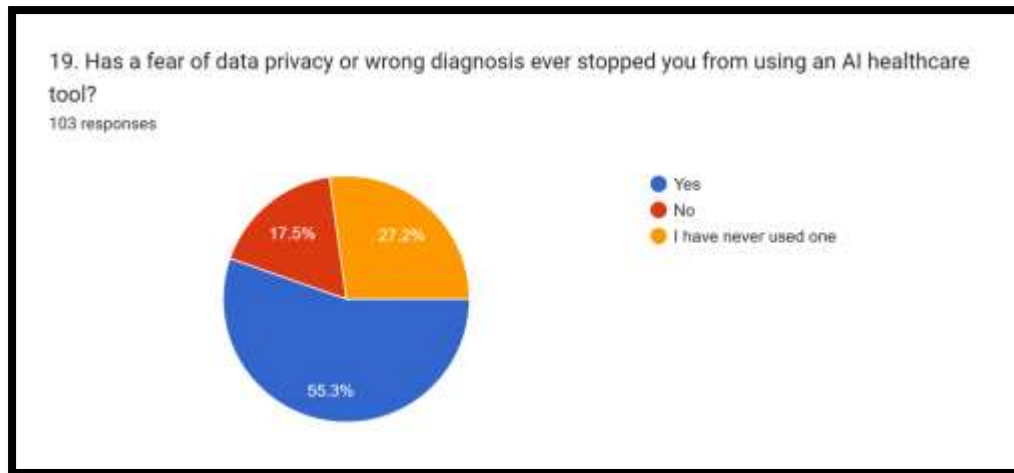
Privacy and Security Concerns

Several respondents reported concerns about:

- Data privacy
- Security risks
- Wrong diagnosis
- Misuse of patient information

Some participants also stated that fear of privacy issues stopped them from using AI healthcare tools.

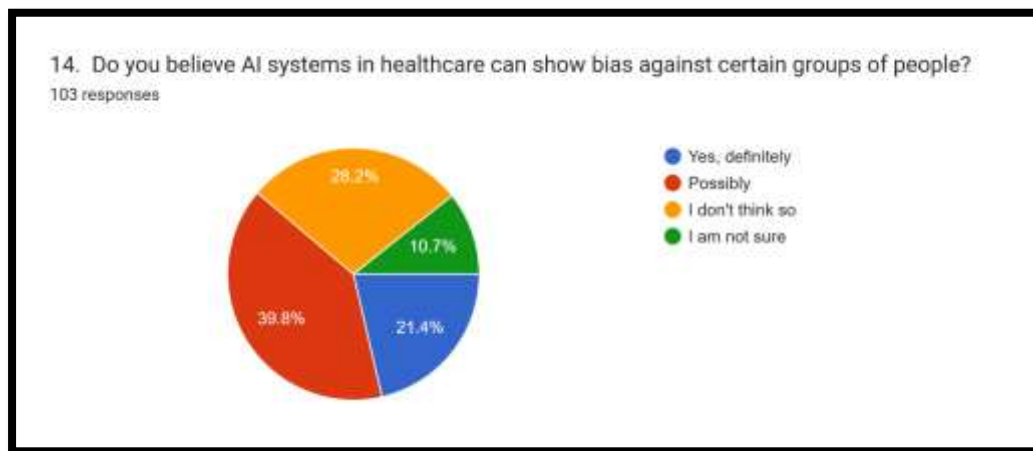




Bias and Ethical Concerns

Many respondents believed that AI systems can become biased if they are trained on incomplete or unfair data. Participants were concerned that AI may not provide equal healthcare recommendations for all patient groups.

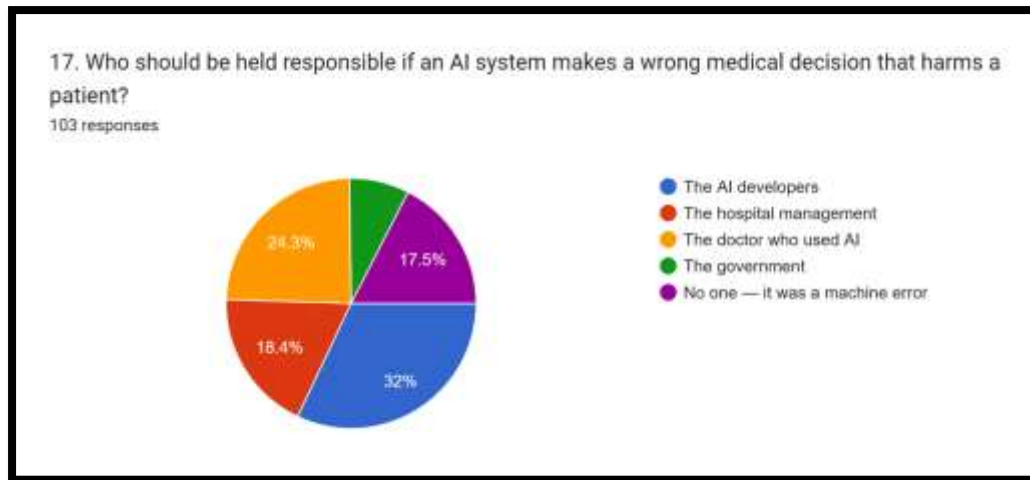
This finding highlights the importance of fairness, diversity in training datasets, and ethical monitoring of AI systems.



Responsibility for AI Mistakes

Survey participants gave different opinions about responsibility for AI errors. Some believed doctors should be responsible, while others blamed hospitals or AI developers.

This shows that accountability in AI healthcare systems is still unclear.

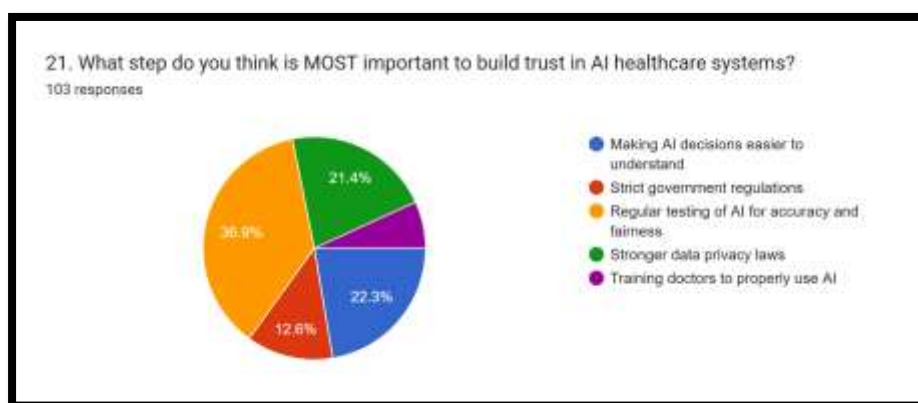


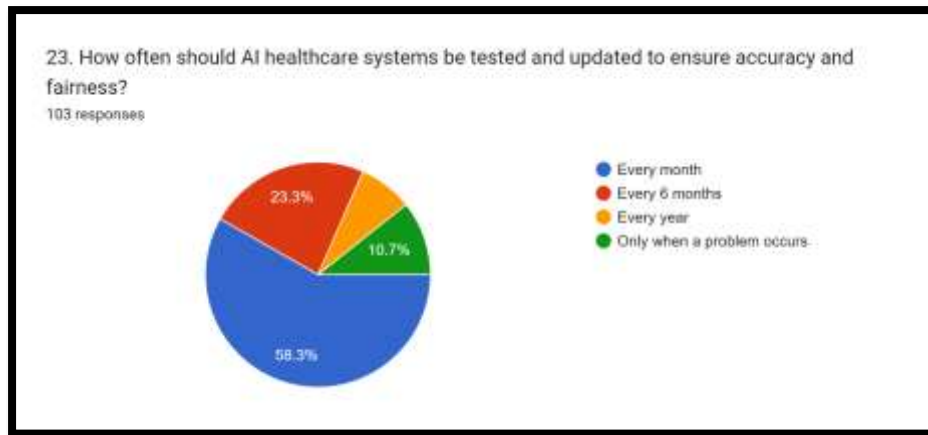
Building Trust in AI Healthcare

Many respondents believed that regular testing of AI systems is the most important step for building trust.

Other important suggestions included:

- Better transparency
- Strong security systems
- Human supervision
- Government regulations
- Proper training for doctors

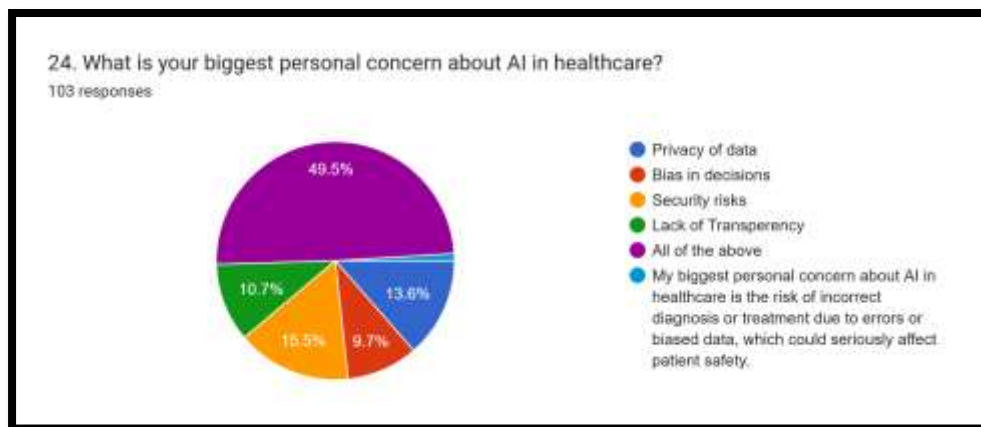




Personal Concerns About AI in Healthcare

Survey participants shared several personal concerns regarding AI in healthcare systems. The most common concerns included wrong diagnosis, misuse of medical data, lack of transparency, and reduced human interaction in healthcare.

These concerns show that people still want emotional support, trust, and human judgment in medical decision-making.



Overall Discussion

The survey results clearly show that people recognize the benefits of AI in healthcare, but ethical and trust-related concerns still exist.

AI can improve healthcare quality, speed, and efficiency, but people do not want complete dependence on machines. Human involvement remains important for safety, empathy, and accountability.

The analysis also highlights the need for explainable AI systems, better privacy protection, and strong ethical guidelines.

7. Ethical Analysis

Ethics is one of the most important aspects of AI in healthcare because healthcare decisions directly affect human life and safety. AI systems must follow professional ethical standards to ensure fairness, safety, transparency, and public trust.

This section analyzes ethical concerns using IEEE, ACM, and BCS Codes of Ethics.

Ethical Analysis Table

Ethical Issue	Mapping with IEEE / ACM / BCS Codes	Analysis	Stakeholder Impact
Patient Privacy & Safety	ACM: Respect Privacy, IEEE: Protect Public Welfare	AI systems collect sensitive patient information. Weak security can lead to data leaks and misuse of medical records.	Patients may lose trust in healthcare organizations.
Incorrect Diagnosis	ACM: Avoid Harm, IEEE: Accept Responsibility	AI systems can sometimes generate inaccurate results, which may affect patient treatment.	Patients can face serious health risks.
Bias and Discrimination	ACM: Be Fair and Avoid Discrimination	AI models trained on biased data may give unfair treatment recommendations.	Certain groups may receive unequal healthcare services.
Lack of Transparency	IEEE: Be Honest and Transparent	Many AI systems do not clearly explain how decisions are made.	Patients and doctors may not trust AI decisions.
Overdependence on AI	BCS: Protect Public Interest	Excessive dependence on AI may reduce human judgment and empathy in healthcare.	Quality of patient care may decrease.
Trust & Reliability	ACM: Maintain Trust, IEEE: Ensure System Reliability	AI systems must provide accurate and consistent results. Unreliable systems can reduce confidence in healthcare technology.	Patients and healthcare professionals may hesitate to use AI systems.
Professional Responsibility	IEEE: Accept Responsibility, BCS: Professional Competence and Integrity	Developers, hospitals, and healthcare professionals must ensure ethical and safe use of AI systems. Negligence may create	Healthcare organizations may face legal action and loss of professional credibility.

Ethical Issue	Mapping with IEEE / ACM / BCS Codes	Analysis	Stakeholder Impact
		serious ethical and legal issues.	

Ethics plays a very important role in healthcare AI systems because these technologies directly affect human lives.

Ethical Issue 1: Patient Privacy

Relevant Ethical Code

- ACM Code: Respect privacy
- IEEE Code: Protect public welfare and safety
- BCS Code: Protect confidential information

Analysis

AI systems require large amounts of patient data for training and decision-making. If patient records are leaked or shared without permission, it can violate privacy rights.

Healthcare organizations must protect sensitive data using secure systems and encryption methods.

Stakeholder Impact

- Patients may lose trust in hospitals.
- Hospitals may face legal action.
- Society may become afraid of healthcare technology.

Ethical Issue 2: Incorrect Diagnosis

Relevant Ethical Code

- ACM Code: Avoid harm
- IEEE Code: Accept responsibility for decisions

Analysis

AI systems are not always perfect. If an AI system gives the wrong diagnosis, it can harm patients or delay treatment.

Doctors should not completely depend on AI systems. Human supervision is necessary to reduce risks.

Stakeholder Impact

- Patients may suffer health problems.
- Doctors may lose credibility.
- Hospitals may face ethical and legal issues.

Ethical Issue 3: Bias and Fairness

Relevant Ethical Code

- ACM Code: Be fair and avoid discrimination
- BCS Code: Act in public interest

Analysis

AI systems can become biased if training data is incomplete or unfair. For example, an AI system trained mainly on one population may not perform equally for all groups.

Developers should use diverse datasets and regularly test systems for fairness.

Stakeholder Impact

- Some patients may receive unfair treatment.
- Public trust may decrease.
- Healthcare inequality may increase.

Ethical Issue 4: Lack of Transparency

Relevant Ethical Code

- IEEE Code: Be honest and transparent
- ACM Code: Maintain trust

Analysis

Many AI systems work without clearly explaining how decisions are made. Patients and doctors may not trust a system they cannot understand.

Explainable AI models should be used in healthcare to improve understanding and confidence.

Stakeholder Impact

- Patients may reject AI systems.
- Doctors may hesitate to use AI tools.
- Decision-making becomes difficult to justify.

Ethical Issue 5: Replacement of Human Doctors

Relevant Ethical Code

- ACM Code: Contribute to human well-being
- BCS Code: Respect professional competence

Analysis

AI systems can support healthcare professionals, but replacing doctors completely may reduce empathy and human interaction in healthcare.

Patients often need emotional support and communication, which machines cannot fully provide.

Stakeholder Impact

- Patients may feel uncomfortable.
- Doctors may fear job replacement.
- Healthcare quality may become less personal.

Ethical Summary

The ethical analysis shows that AI healthcare systems must focus on safety, fairness, transparency, and human-centered care. Ethical principles from IEEE, ACM, and BCS provide important guidance for responsible use of AI.

8. Legal and Regulatory Issues

AI healthcare systems must follow legal and regulatory standards because they process highly sensitive patient information and support critical medical decisions.

PECA (Pakistan Electronic Crimes Act)

Pakistan Electronic Crimes Act (PECA) addresses cybercrime, unauthorized access, and misuse of digital systems.

If healthcare databases are hacked or patient information is leaked, healthcare organizations may face legal consequences under cybercrime laws.

The survey responses also showed that many participants fear misuse of personal medical information. This highlights the importance of stronger cybersecurity systems in healthcare organizations.

GDPR and Data Protection

Although GDPR is a European regulation, it is considered an international standard for privacy and data protection. GDPR emphasizes:

- User consent
- Transparency
- Right to privacy
- Secure data handling
- Accountability

Healthcare AI systems should follow similar standards to protect patient rights.

Intellectual Property Rights (IPR)

AI software, healthcare applications, and medical algorithms are protected through intellectual property laws. Developers must avoid unauthorized use of medical records, datasets, and software systems.

Liability and Accountability Issues

One of the biggest legal challenges is determining responsibility when AI systems make incorrect decisions. For example:

- If AI gives a wrong diagnosis, should the doctor be blamed?
- Should responsibility fall on hospitals?
- Should software companies or AI developers be legally responsible?

Current legal systems still lack clear answers for many AI-related healthcare situations.

Compliance Gaps and Risks

Many healthcare organizations adopt AI systems quickly without proper ethical testing or regulation. This creates risks such as:

- Data breaches
- Biased medical decisions
- Unsafe recommendations
- Patient discrimination
- Loss of public trust

Critical Legal Analysis

Current healthcare laws are still developing and are not fully prepared for advanced AI systems. Technology is growing faster than regulations, creating gaps in accountability and safety standards. Without strong regulation, healthcare organizations may prioritize efficiency and profit over patient safety and ethics. Therefore, governments must create specialized AI healthcare regulations that focus on transparency, fairness, safety, and human rights.

9. Recommendations / Proposed Solution

The survey findings and ethical analysis show that AI in healthcare can provide major benefits, but only when it is implemented responsibly. The following recommendations can improve trust, ethics, safety, and public acceptance of healthcare AI systems.

1. AI Should Support Doctors, Not Replace Them

AI systems should be used as decision-support tools instead of replacing healthcare professionals completely. Human supervision is necessary because doctors provide:

- Professional judgment
- Communication
- Emotional support
- Ethical decision-making

This recommendation directly supports ACM and BCS ethical principles related to public welfare and human-centered care.

2. Improve Transparency Through Explainable AI

Healthcare organizations should use explainable AI systems that clearly show how decisions are made. Patients and doctors are more likely to trust systems that provide understandable explanations. Transparency also supports IEEE ethical principles related to honesty and accountability.

3. Strengthen Data Privacy and Cybersecurity

Hospitals should implement:

- Encrypted databases
- Access control systems
- Multi-factor authentication
- Cybersecurity monitoring
- Regular security audits

These measures can reduce the risk of healthcare data breaches.

4. Reduce Bias Through Diverse Training Data

AI systems should be trained using diverse datasets representing different genders, age groups, and communities. Regular fairness testing should also be performed to identify biased outcomes.

5. Establish Clear Legal Accountability

Governments should define clear rules about responsibility for AI-related healthcare mistakes. This can reduce confusion among patients, doctors, hospitals, and software companies.

6. Ethical Training for Professionals

Developers and healthcare professionals should receive training related to:

- AI ethics
- Privacy protection
- Responsible AI usage
- Professional responsibilities

This will help professionals make safer and more ethical decisions.

7. Government Regulation and Certification

Governments should create specialized regulatory frameworks for healthcare AI systems. AI systems should be tested and certified before being used in hospitals.

Implementation Plan

Step	Action	Responsible Stakeholder
1	Develop ethical AI policies	Government & Hospitals
2	Implement cybersecurity systems	IT Teams
3	Conduct fairness and safety testing	AI Developers
4	Train healthcare professionals	Hospitals & Universities
5	Monitor AI performance regularly	Regulatory Authorities

Critical Recommendation

AI should never be treated as a completely independent replacement for healthcare professionals. Technology can improve efficiency, but healthcare also requires empathy, ethics, communication, and human understanding. Responsible AI means balancing innovation with patient safety and public trust.

10. Conclusion

AI has the potential to improve healthcare systems by increasing speed, efficiency, and accuracy. It can support doctors in diagnosis, treatment planning, and patient management. However, trust and ethical concerns remain major challenges. Survey results show that people are interested in AI healthcare technologies, but they still worry

about privacy, transparency, wrong diagnosis, and accountability. The report also highlights the importance of ethical principles from IEEE, ACM, and BCS codes. AI systems must be fair, transparent, secure, and supervised by humans. Overall, AI should be used as a support tool to improve healthcare rather than replace healthcare professionals completely. Responsible development, strong regulations, and ethical practices are necessary to build public trust.

11. Personal Reflection

This project helped us understand the importance of ethics and professionalism in computer science. Before this project, I mainly focused on the technical side of AI, but now I understand that technology also affects society, privacy, safety, and human trust. We learned how AI can create both opportunities and challenges in healthcare systems. I also learned how ethical codes such as ACM, IEEE, and BCS guide professionals in making responsible decisions. One of the main challenges during this project was analyzing survey data and connecting it with ethical theories. It was also difficult to understand legal issues related to healthcare technology. We improved our research, communication, and analytical skills through this report. This project also increased our awareness about professional responsibilities as a future computing professional.

12. References

- **Survey Link:**
<https://docs.google.com/forms/d/e/1FAIpQLSeo1Zu8iaUNzXONG6b9iKxKDkNgkmtrC1k5jq4HDH6zmEVLwg/viewform?usp=publish-editor>
- Baase, S. “A Gift of Fire: Social, Legal, and Ethical Issues for Computing Technology.”
- Johnson, D. “Computer Ethics.”
- ACM Code of Ethics and Professional Conduct.
- IEEE Code of Ethics.
- BCS Code of Conduct.
- World Health Organization (WHO) reports on AI in healthcare.
- Research articles on AI ethics and healthcare systems.
- GDPR Data Protection Guidelines.
- Pakistan Electronic Crimes Act (PECA).
- Healthcare AI journal articles and survey reports.

13. Appendices

Appendix A – Survey Questionnaire

The survey included questions related to:

- Trust in AI diagnosis
- Privacy concerns
- Transparency of AI systems
- AI-based treatment acceptance
- Responsibility for AI mistakes
- Suggestions for safer AI healthcare systems

Appendix B – Survey Charts and Graphs

Charts created from survey responses include:

- Age group distribution
- Trust in AI systems
- Acceptance of AI treatment
- Privacy concerns
- Transparency opinions
- Responsibility for AI errors

Appendix C – Survey Findings Summary

Key findings from the survey:

- Most participants support AI with doctor supervision.
- Privacy and security are major concerns.
- Transparency is important for trust.
- AI should support doctors instead of replacing them.
- Regular testing is necessary for safe healthcare AI.